

Simulation Output Limitation Module (SOLM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Operational Layer: Simulation Output Governance Layer

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Subcategory: Simulation Governance

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Governed Space: Simulation Output Limitation

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Simulation Output Limitation

Canonical Definition

Simulation Output Limitation Module (SOLM) defines the governance framework through which material limitations attached to Simulation Outputs are identified, classified, bounded, documented, propagated, and preserved so that outputs cannot be interpreted, generalized, transferred, compared, or relied upon beyond the conditions actually supported by the simulation that produced them.

SOLM governs the boundary between:

what a Simulation Output shows

and:

what that output does not legitimately establish.

Its purpose is to prevent computational precision, high simulation volume, visually convincing outputs, successful scenario execution, or apparently stable results from expanding the legitimate meaning of a Simulation Output beyond its governed simulation basis.

Operational Role

SOLM serves as the fourth internal governance space of the Simulation Output Governance Standard (SOGS).

The preceding modules establish:

SOIM — What exactly did the simulation produce?

SOCM — What type of Simulation Output is it?

SIM — What inference, if any, may legitimately be drawn from that output?

SOLM asks:

Where does the legitimate meaning of that output stop, what conditions constrain its interpretation, and which limitations must remain attached whenever the output is communicated, compared, reused, or relied upon?

The governed progression is:

Identified Simulation Output

↓

Output Classification

↓

Governed Inference

↓

Limitation Identification

↓

Limitation Materiality

↓

Interpretation Boundary

↓

Applicability Restriction

↓

Transfer / Generalization Restriction

↓

Limitation Propagation



Bounded Simulation Output



Output Documentation

Module Operational Space

SOLM governs: Simulation Output limitation identification, limitation source, limitation classification, limitation materiality, output applicability boundaries, scenario-bound limitations, model-bound limitations, environment-bound limitations, input-bound limitations, assumption-bound limitations, parameter-bound limitations, execution-bound limitations, observation-bound limitations, statistical limitations, uncertainty limitations, resolution limitations, temporal limitations, spatial limitations, population limitations, configuration limitations, version limitations, edge-case limitations, stress-condition limitations, scenario-coverage limitations, inference limitations, generalization limitations, transfer limitations, comparison limitations, reuse limitations, extrapolation limitations, aggregation limitations, communication limitations, decision-use limitations, unresolved limitation, limitation propagation, limitation reassessment, limitation versioning, and Output Limitation Traceability.

Module Function

SOLM exists to prevent: scenario-specific results being presented as universal results, model-specific behavior being presented as real-world certainty, nominal-condition outputs being generalized to edge or stress conditions, outputs produced under one environment being applied to another without qualification, outputs based on one population being generalized to another, simulated time horizons being extended beyond what was modeled, stochastic results being represented as deterministic truth, point estimates being presented without material uncertainty, averages hiding adverse distributions, incomplete Scenario Coverage being forgotten, unknown Scenario Space disappearing from downstream interpretation, simulation capability limits being mistaken for system limits, inferred results being presented as direct observations, correlation being presented as causation, output precision being confused with output certainty, and output reuse occurring after the governing simulation conditions have materially changed.

Simulation Output Limitation

A Simulation Output Limitation is any material condition restricting what a Simulation Output may legitimately: describe, imply, represent, compare, generalize, predict, support, transfer, or be relied upon to establish.

A limitation may arise from any upstream SIMULOS® governance space.

Limitation Origin

A material output limitation should remain traceable to its origin where possible.

Possible origins include:

Simulation Object

Simulation Purpose & Scope

Reality Representation

Simulation Assumptions

Model & Environment

Simulation Inputs & Data

Scenario Design

Simulation Execution

Output Classification

or:

Inference Process

Conceptually:

Upstream Limitation → Simulation Output Limitation Upstream Limitation
Propagation

SIMULOS® establishes:

A material limitation does not disappear merely because the simulation successfully produced an output.

If a limitation exists upstream and materially affects what the result can mean, it should propagate into Output Governance.

Output Applicability Boundary

An Output Applicability Boundary defines where a Simulation Output may legitimately apply.

Applicability may depend upon: Simulation Object version, system configuration, population, environment, geography, time period, operating state, scenario family, parameter range, input conditions, autonomy state, or another relevant boundary.

Scope-Bound Output

A Simulation Output should remain bounded by the Simulation Purpose & Scope under which it was produced.

For example:

simulation of braking behavior at 0–80 km/h

does not automatically support conclusions about:

120 km/h behavior.

No Scope Expansion Principle

SIMULOS® establishes:

A Simulation Output must not acquire a broader scope than the simulation inquiry that produced it unless that broader applicability is independently established.

Reality Representation Limitation

Outputs are constrained by what the simulation represented.

If Reality Representation excluded a material real-world feature, the resulting Output cannot legitimately establish behavior involving that omitted feature.

Therefore:

Unrepresented Reality → Unsupported Output Claim Abstraction Limitation

A simulation may deliberately simplify reality.

An output derived from that abstraction should not be presented as if the omitted detail had been simulated.

Fidelity Limitation

Simulation fidelity may be sufficient for one purpose but insufficient for another.

For example:

a low-fidelity simulation may support early design exploration while remaining unsuitable for detailed safety reliance.

SOLM preserves that boundary.

Assumption-Bound Limitation

Simulation Outputs may depend upon assumptions established under SAS.

Where an output materially depends upon an assumption, its interpretation should remain conditional upon that assumption or upon the range within which

it remains applicable.

Conceptually:

Output Validity of Interpretation $\leq$ Applicability of Material Assumption No
Assumption Erasure Principle

A simulation assumption must not disappear from output interpretation when the output materially depends upon that assumption.

Model-Bound Limitation

A Simulation Output reflects behavior of the governed simulation model.

It does not automatically establish equivalent behavior of the real system.

This distinction is fundamental:

Model Behavior $\neq$ Operational Reality by Default Environment-Bound Limitation

The simulation environment may constrain the legitimate meaning of the output.

A result produced under:

Environment Version A

may not automatically transfer to:

Environment Version B

where material conditions differ.

Configuration Limitation

Outputs may apply only to a specific: hardware configuration, software version, model version, sensor package, mission configuration, autonomy level, or operational configuration.

Input-Bound Limitation

Simulation Outputs cannot legitimately exceed the representativeness, temporal applicability, transformation basis, and uncertainty of the Simulation Inputs on which they materially depend.

For example:

Input Population A

does not automatically support:

Output Claim for Population B.

Representativeness Limitation

Where SIDS identifies incomplete representativeness, that limitation should remain attached to the output.

A simulation using input data representative only of:

urban daytime driving

cannot silently produce an output interpreted as representative of:

all driving environments.

Temporal Input Limitation

Historical or temporally restricted inputs may bound Output applicability.

For example:

an economic simulation using pre-regime-change data may not automatically support post-regime conclusions.

Input Uncertainty Limitation

Material uncertainty in the Simulation Input basis should constrain downstream interpretation.

A precise output derived from uncertain inputs may remain materially uncertain.

No Precision Laundering Principle

SIMULOS® establishes:

A simulation may transform uncertain inputs into computationally precise outputs, but it may not transform uncertainty into certainty merely through computation.

Scenario-Bound Limitation

Outputs may apply only to the Scenario or Scenario Family under which they were generated.

A nominal-scenario Output does not automatically establish behavior under: edge cases, stress conditions, degraded states, failure conditions, transitions, or recovery scenarios.

Scenario Coverage Limitation

A Simulation Output may be locally strong while the overall Scenario Set remains incomplete.

SCCM may identify:

Scenario Coverage Gap

or:

Critical Scenario Coverage Gap.

SOLM ensures that such gaps constrain any broader Output claim.

No Coverage Inflation Principle

SIMULOS® establishes:

Strong results inside covered Scenario Space do not establish behavior inside uncovered Scenario Space.

Unknown Scenario Space Limitation

Where unknown Scenario Space remains material, Output interpretation should avoid implying exhaustive behavioral knowledge.

Edge-Case Limitation

Outputs produced under nominal or ordinary variation should not be interpreted as demonstrating behavior at difficult edges unless those edges were explicitly examined.

Stress-Condition Limitation

Successful performance under ordinary conditions does not establish robustness under stress.

Likewise, one stress condition does not establish robustness across all stress combinations.

Execution-Bound Limitation

A Simulation Output inherits relevant limitations from the execution that produced it.

Examples include: execution deviation, interrupted run, incomplete run, unexpected intervention, configuration mismatch, resource limitation, execution drift, or incomplete observation.

Execution Completeness Limitation

An output from a partially completed execution should not be represented as if it came from a complete execution where completion materially affects interpretation.

Observation Limitation

SES may identify Observation Blind Spots.

Where material internal events or states were not observable, Output

interpretation should not imply knowledge about those hidden regions.

Randomness Limitation

Where simulation behavior is stochastic, one run may not characterize the distribution of possible outcomes.

SOLM may therefore preserve a limitation such as:

Single stochastic realization — not sufficient to characterize outcome distribution.

Repetition Limitation

Repeated runs may remain insufficient where: repetitions are too few, configuration drift exists, seed coverage is narrow, relevant variation is missing, or runs are not comparable.

Statistical Limitation

Simulation Outputs may have statistical limitations involving: sample size, variance, confidence, distribution tails, sampling strategy, rare-event representation, convergence, or estimator uncertainty.

Mean Limitation

Average behavior may conceal: extremes, variance, multimodality, adverse tails, or subgroup differences.

SIMULOS® establishes:

An average Simulation Output does not represent every simulated condition or every possible outcome.

Distribution Limitation

A distribution estimated from simulation is bounded by: scenario generation, sampling method, assumptions, model behavior, input distributions, and number of executions.

It is not automatically equivalent to the real-world distribution.

Rare-Event Limitation

Low observed frequency in simulation may result from: genuine rarity, inadequate sampling, scenario-selection bias, insufficient run count, or missing Scenario Space.

A rare simulated event rate should therefore not automatically be interpreted as a reliable real-world probability.

Resolution Limitation

Output resolution may constrain interpretation.

Examples include: temporal resolution, spatial resolution, state resolution, population granularity, behavioral granularity, or aggregated outputs.

No Resolution Inflation Principle

A Simulation Output cannot legitimately reveal distinctions that the simulation did not represent or observe at sufficient resolution.

Temporal Output Limitation

A Simulation Output may apply only within a defined simulated time horizon.

For example:

0–24 hours

does not automatically support:

30-day behavior.

Long-Horizon Limitation

Uncertainty, model drift, state accumulation, or changing environments may increase with simulated horizon.

Outputs should remain bounded accordingly.

Spatial Limitation

A simulation covering one: location, terrain, orbital region, climate, infrastructure topology, or geographic region

does not automatically support spatial generalization elsewhere.

Population Limitation

Outputs may be bounded to the population represented by: data, actor models, simulation scenarios, or relevant configuration.

Version Limitation

Simulation Outputs are attached to the versions that produced them.

Conceptually:

Output O1 → Model v3.1 + Environment v5.0 + Input Set 22 + Scenario Set 7

A later model or environment version should not automatically inherit the Output.

No Version Inheritance Principle

SIMULOS® establishes:

Simulation Outputs do not automatically remain applicable after a material change in the Simulation Object, Model, Environment, Inputs, Parameters, or Scenario Basis.

Inference Limitation

SIM — Simulation Inference Module establishes what may legitimately be inferred from a Simulation Output.

SOLM establishes the boundary beyond which that inference must not expand.

Direct Output vs Inferred Meaning

For example:

Direct Output: simulated component temperature reached 112°C.

Possible inference:

Under the simulated conditions, the component model crossed the defined thermal threshold.

Unsupported inference without further basis:

The real component will fail at exactly 112°C in all operational environments.

SOLM governs that distinction.

No Observation-to-Causation Principle

A simulated association does not automatically establish causal truth.

If:

Condition A

and:

Output B

occur together in simulation, causality requires a sufficient simulation and inference basis.

No Simulation-to-Reality Equivalence Principle

SIMULOS® establishes:

What occurred in simulation is evidence of simulated behavior under governed simulated conditions; it is not automatically proof of identical behavior in Operational Reality.

ORA™ and VALIDOS® may participate in downstream correlation and validation.

Generalization Limitation

A Generalization Limitation restricts extension of an output beyond the represented conditions.

Generalization may concern: population, environment, geography, time, configuration, system type, behavior, or operating condition.

Transfer Limitation

A Simulation Output generated for one Simulation Object should not automatically be transferred to another object merely because the objects are similar.

Cross-System Transfer

For example:

Robot Model A

may share architecture with:

Robot Model B

but differences in: mass, software, sensors, actuators, control logic, or environment

may materially limit transfer.

Comparison Limitation

Two Simulation Outputs should not be compared as equivalent where material differences exist in: Simulation Purpose, Scenario, Model, Environment, Inputs, Parameters, execution conditions, output type, or inference basis.

No Apples-to-Oranges Principle

SIMULOS® establishes:

Numerically similar Simulation Outputs are not necessarily methodologically comparable.

Reuse Limitation

A Simulation Output may be reused in: another analysis, another scenario, another model, another validation process, another decision, or another simulation.

Reuse should remain bounded by the Output's original applicability and limitations.

No Context-Free Reuse Principle

A Simulation Output must not be detached from the conditions necessary to understand what it legitimately means.

Extrapolation Limitation

Outputs should not be extrapolated outside: modeled ranges, simulated horizons, Scenario Space, input distributions, or represented boundaries

without explicit governance.

Aggregation Limitation

Aggregated Simulation Outputs may hide important: scenario differences, subgroup differences, tail behavior, failures, or contradictory outcomes.

Output Selection Limitation

If only selected outputs or successful runs are presented, interpretation may become biased.

Material negative, anomalous, or contradictory outputs should remain visible where they affect the legitimate conclusion.

No Cherry-Picked Output Principle

SIMULOS® establishes:

A Simulation Output claim must not be strengthened by selectively excluding materially relevant outputs that contradict or weaken that claim.

Output Contradiction

Simulation Outputs may conflict across: scenarios, repetitions, versions, parameter configurations, or model states.

A contradiction may itself create an Output Limitation.

Unresolved Contradiction Limitation

Where conflicting outputs cannot yet be reconciled, the legitimate inference may need to remain:

Conditional

Partial

Conflicted

or:

Undetermined.

SOLM preserves the limitation.

Decision-Use Limitation

A Simulation Output may support:

exploration

without supporting:

deployment authorization.

It may support:

design comparison

without supporting:

safety certification.

It may support:

hypothesis generation

without supporting:

operational reliance.

The intended decision context matters.

Exploratory Output

An exploratory Simulation Output may legitimately support: investigation, hypothesis generation, design direction, anomaly discovery, or scenario refinement.

It should not automatically be represented as validated evidence for high-consequence reliance.

Operational Reliance Limitation

SOGS does not authorize operational reliance.

VALIDOS®, ORA™, ASGA™, GOA™, or other relevant architectures may determine what downstream reliance is legitimate.

SOLM preserves limitations that those downstream processes need to know.

Communication Limitation

Limitations should remain visible when Simulation Outputs are: reported, visualized, summarized, presented to management, published, transferred between teams, used by AI systems, or integrated into decision-support tools.

No Limitation Loss Through Summarization Principle

SIMULOS® establishes:

A material Output Limitation must not disappear merely because the underlying simulation record is summarized into a dashboard, chart, score, executive summary, or AI-generated explanation.

Machine-Readable Limitation

Where Simulation Outputs are consumed automatically, limitations may require machine-readable representation.

Examples may include: applicability tags, scenario constraints, object-version references, uncertainty flags, scope boundaries, comparison restrictions, or reliance warnings.

Limitation Classification

A material Output Limitation may be classified as:

Scope Limitation

Representation Limitation

Assumption Limitation

Model Limitation

Environment Limitation

Input Limitation

Scenario Limitation

Coverage Limitation

Execution Limitation

Observation Limitation

Statistical Limitation

Uncertainty Limitation

Temporal Limitation

Spatial Limitation

Population Limitation

Configuration Limitation

Version Limitation

Inference Limitation

Generalization Limitation

Transfer Limitation

Comparison Limitation

Reuse Limitation

or another simulation-specific class.

Limitation Materiality

Not every known limitation requires equal prominence.

A limitation is material where ignoring it could reasonably change: interpretation, inference, comparison, decision, validation, operational reliance, or assessment of the Simulation Object.

Critical Output Limitation

A Critical Output Limitation exists where ignoring the limitation could produce materially misleading or high-consequence interpretation.

Limitation Severity

A simulation implementation may classify limitations as:

Informational

Material

Major

Critical

or another justified structure.

SIMULOS® does not mandate one universal severity scale.

Limitation Status

Possible statuses may include:

Active

Conditionally Resolved

Resolved

Superseded

Unresolved

or:

Applicability Undetermined.

Resolved Limitation

A limitation may be resolved only when an appropriate governed process establishes that the restriction no longer materially applies.

For example:

an earlier population limitation may be resolved after additional simulations extend the represented population sufficiently.

No Silent Resolution Principle

SIMULOS® establishes:

A Simulation Output Limitation must not be treated as resolved merely because later results are favorable.

Limitation Propagation

A material limitation should follow the Output into downstream use.

Conceptually:

Simulation Output

↓

Output Limitation

↓

Inference

↓

Documentation

↓

Validation / Analysis / Decision / Reliance

The limitation should remain visible wherever its absence could materially distort meaning.

Limitation Propagation Failure

A Limitation Propagation Failure occurs where the original simulation record contains a material limitation but downstream reporting or use no longer communicates it.

Examples include: dashboard removes uncertainty, executive summary removes scope restriction, reused data loses Scenario reference, AI-generated answer omits model limitation, or published graph removes excluded conditions.

Output Limitation Register

SOLM may maintain a Simulation Output Limitation Register.

A limitation record may identify: Limitation ID, affected Output, limitation class, origin, description, materiality, applicability, affected inference, affected comparison, downstream propagation requirement, status, version, and reassessment trigger.

Limitation-to-Output Binding

Every material limitation should be bound to the relevant: individual Output, Output Set, Scenario Output, Campaign Output, derived result, or inference.

Limitation-to-Inference Binding

Where a specific limitation constrains a particular inference:

Output → Inference → Limitation

should remain traceable.

Limitation-to-Decision Binding

Where an Output is used for decision support, material limitations should remain available to that decision process.

Limitation Reassessment Trigger

Reassessment may be required when: additional scenarios are executed, coverage expands, new data becomes available, model fidelity improves, environment changes, assumptions change, input uncertainty decreases, new validation evidence appears, the Output is reused for another purpose, or a new reliance context is proposed.

Output Limitation Change

A limitation may: narrow, expand, become more severe, become less severe, resolve, or become newly discovered

as the simulation lifecycle evolves.

Historical Limitation Preservation

Later resolution of a limitation must not rewrite the limitation state of earlier Output use.

Point-in-time reconstruction should remain possible.

No Retrospective Strengthening Principle

SIMULOS® establishes:

A later improvement in simulation capability must not retroactively strengthen what an earlier Simulation Output legitimately established at the time it was produced.

Point-in-Time Output Interpretation

SOLM should support questions such as:

What limitations were known when this Output was produced?

Which assumptions constrained it?

Which Scenario Space was uncovered?

Which model version produced it?

Which inference was allowed?

Which generalizations were prohibited?

Was the limitation later resolved?

Was the Output reused outside its original context?

Simulation Output Boundary Statement

A governed Output may carry a concise boundary statement such as:

This Simulation Output applies only to the identified Simulation Object version, Simulation Scope, Scenario Set, Model & Environment configuration, Input & Data Basis, execution conditions, and Output limitations recorded for the producing simulation activity.

Space and Mission Systems

Space and mission simulation creates particularly important Output Limitation questions because operational reality may be difficult or impossible to reproduce before deployment.

A mission simulation may produce:

95.8% successful autonomous recovery

across a large Monte Carlo campaign.

That number alone does not establish:

95.8% real-mission recovery probability.

SOLM asks:

Which mission conditions were represented?

Which communication states were absent?

Which hardware failures were modeled?

Which dependency combinations remained uncovered?

Which environmental uncertainties were simplified?

Was the recovery model validated against comparable operational behavior?

The Output may remain extremely useful.

But its legitimate meaning must remain bounded by the simulated conditions.

Autonomous Systems

An autonomous system may achieve:

99.9% simulated task success.

SOLM prevents that number from silently becoming:

99.9% operational safety

or:

99.9% real-world reliability

unless the required governance basis independently supports those claims.

The Output might legitimately establish:

99.9% success within the defined governed Scenario Set and simulation conditions.

That is a stronger methodological statement because its boundary is explicit.

AI Agent Systems

An AI agent may complete:

98% of simulated workflows

within a test environment.

SOLM may reveal that: tool permissions were fixed, external systems were stable, memory was always available, human instructions were unambiguous, and authority transitions were not simulated.

The legitimate Output claim therefore remains bounded to that condition space.

Industrial Simulation

A manufacturer may simulate equipment under:

normal + elevated load

and observe no failures.

SOLM prevents that result from being generalized automatically to: extreme temperature, degraded maintenance, sensor failure, long-duration operation, or combined load and environmental stress

if those conditions were not represented.

Existing Simulation Program Diagnostic

SOLM can be applied to existing simulation programs by asking:

What claims are being made from our Simulation Outputs?

Which output limitations are explicit?

Which limitations exist upstream but disappear from reporting?

Do dashboards show precision without uncertainty?

Are nominal outputs being generalized to uncovered Scenario Space?

Are outputs from old Model Versions still being reused?

Are Monte Carlo frequencies being presented as real-world probabilities without qualification?

Are outputs being used for decisions beyond the purpose for which the simulation was designed?

Can each output claim be traced to its limiting conditions?

This can reveal that the key weakness is not the quality of the simulation itself but the way its outputs are interpreted and communicated.

Module Boundary

SOLM begins when a Simulation Output has been sufficiently: identified under SOIM, classified under SOCM, and associated with a governed inference basis under SIM.

SOLM ends when the materially relevant limitations affecting: interpretation, inference, applicability, generalization, comparison, transfer, reuse, and downstream reliance

have been sufficiently identified, classified, bounded, documented, and attached to the affected Simulation Output.

SOLM does not govern: general integrity, data integrity, output tamper

resistance, validation determination, operational deployment authorization, general risk management, or final reliance authorization.

Integrity remains under INTEGROS®.

Architectural Boundary

SOLM specifically avoids the concept:

Output Integrity

because that governance space belongs to INTEGROS®.

The architectural distinction is:

SOLM → What limitations constrain the meaning and use of this Simulation Output?

INTEGROS® → Has the relevant Output or record preserved the integrity properties required of it?

This separation prevents SIMULOS® from duplicating integrity governance.

Minimum Implementation Framework

1. Identify Material Output Limitations

Determine which upstream or output-specific conditions materially constrain interpretation. 2. Classify Limitation Origin

Identify whether the limitation arises from: scope, representation, assumption, model, environment, inputs, scenarios, execution, observation, statistics, uncertainty, or inference.

3. Assess Limitation Materiality

Determine whether ignoring the limitation could materially alter interpretation, comparison, validation, or reliance. 4. Establish Output Applicability Boundary

Define where the Output may and may not legitimately apply. 5. Establish Inference Boundary

Determine what the Output may support and which broader conclusions remain unsupported. 6. Establish Generalization and Transfer Restrictions

Identify restrictions upon use across: populations, environments, versions, systems, scenarios, time periods, and configurations.

7. Bind Limitations to Outputs

Ensure each material limitation remains attached to the affected Output or Output Set. 8. Establish Propagation Requirements

Determine where limitations must remain visible in downstream: documentation, reporting, validation, analysis, decision support, and reuse.

9. Define Reassessment Triggers

Identify which new simulations, evidence, model changes, Scenario expansions, or reuse contexts require limitation reassessment. 10. Preserve Historical Limitation State

Maintain the limitation record applicable at the time the Output was produced and used.

Governance Outputs

SOLM may produce: Simulation Output Limitation Identifier, Simulation Output Limitation Record, Output Applicability Boundary, Scope Limitation Record, Reality Representation Limitation Record, Abstraction Limitation Record, Fidelity Limitation Record, Assumption-Bound Limitation Record, Model-Bound Limitation Record, Environment-Bound Limitation Record, Configuration Limitation Record, Input-Bound Limitation Record, Representativeness Limitation Record, Temporal Input Limitation Record, Input Uncertainty Limitation Record, Scenario-Bound Limitation Record, Scenario Coverage Limitation Record, Unknown Scenario Space Limitation Record, Edge-Case Limitation Record, Stress-Condition Limitation Record, Execution-Bound Limitation Record, Execution Completeness Limitation Record, Observation Limitation Record, Randomness Limitation Record, Repetition Limitation Record, Statistical Limitation Record, Distribution Limitation Record, Rare-Event Limitation Record, Resolution Limitation Record, Temporal Output Limitation Record, Spatial Limitation Record, Population Limitation Record, Version Limitation Record, Inference Limitation Record, Generalization Limitation Record, Transfer Limitation Record, Comparison Limitation Record, Reuse Limitation Record, Extrapolation Limitation Record, Aggregation Limitation Record, Output Contradiction Limitation Record, Decision-Use Limitation Record, Critical Output Limitation, Output Limitation Register, Limitation Propagation Requirement, Limitation Propagation Failure Record, Limitation Reassessment Trigger, Limitation Change Record, Point-in-Time Limitation Record, Output Boundary Statement, and Simulation Output Limitation Traceability Map.

Use Case 1 — Spacecraft Autonomous Landing Simulation

Scenario

A spacecraft landing simulation produces a result showing:

99.4% successful autonomous landing

across a large governed Simulation Campaign.

The number appears highly convincing. Application

SOLM identifies material limitations: only the modeled terrain classes were represented, communication-loss conditions were limited, certain sensor-failure combinations remained outside Scenario Coverage, atmospheric uncertainty was bounded to a defined range, the Output applies to one vehicle

configuration, and the result is a simulated campaign frequency, not a direct empirical mission-success probability.

Result

The legitimate Output statement becomes:

99.4% simulated landing success across the governed Scenario Set, Model & Environment configuration, Input basis, and uncertainty conditions represented by the identified Simulation Campaign.

The limitation does not weaken the simulation.

It prevents a strong simulation result from being converted into a stronger real-world claim than the simulation can legitimately support. Use Case 2 — Autonomous Vehicle Safety Simulation

Scenario

An autonomous-driving platform reports:

99.99% collision-free simulation performance. Application

SOLM identifies that most Simulation Executions occur within: mapped urban environments, defined weather ranges, supported sensor configurations, known road-user classes, and a bounded Scenario Set.

Some compound edge conditions remain uncovered. Result

The organization cannot legitimately communicate the Output as:

99.99% real-world safety.

The governed Output remains:

99.99% collision-free performance within the identified simulated conditions and Scenario Coverage basis.

The remaining limitations become explicit inputs to further Scenario expansion, validation, and operational reliance assessment.

Architectural Position

Simulation Output Limitation Module (SOLM) is the fourth internal governance space of the Simulation Output Governance Standard (SOGS).

The complete SOGS progression is:

Simulation Output Identification

↓

Simulation Output Classification

↓

Simulation Inference

↓

Simulation Output Limitation

↓

Simulation Output Documentation

SOIM establishes:

What exactly was produced?

SOCM establishes:

What kind of Simulation Output is it?

SIM establishes:

What may legitimately be inferred from it?

SOLM establishes:

Where does that legitimate interpretation stop?

SODM then preserves:

the Output, its classification, inference, limitations, provenance, and documentation basis.

Foundational Principle

A Simulation Output is only as broadly meaningful as the governed conditions under which it was produced, and no output may legitimately escape the assumptions, representation, data, scenarios, execution conditions, uncertainty, and coverage boundaries that materially constrain what it can demonstrate.

Canonical Closing Statement

Simulation Output Limitation Module establishes the boundary-governance layer of the Simulation Output Governance Standard by ensuring that the meaning of Simulation Outputs cannot silently expand beyond the conditions the simulation actually represented and executed. Through explicit limitation origin, materiality, applicability boundaries, assumption, model, environment, input, Scenario Coverage, execution, observation, statistical, temporal, spatial, population, version, inference, generalization, transfer, comparison, reuse, and decision-use limitations, SOLM preserves the distinction between computational output and legitimate conclusion. By propagating material limitations into downstream documentation, validation, analysis, communication, and reliance while explicitly leaving Output Integrity to INTEGROS®, SOLM ensures that strong simulation results remain useful precisely

because their boundaries remain visible, traceable, and governable.

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